REFERENCE CHAPTER: SUPPLY CHAIN NETWORK STRATEGY - PART 1

Centralized vs Decentralized Distribution Networks:

Centralized warehousing maximizes inventory risk pooling and minimizes total facility overhead.

However, centralization exposes companies to long customer delivery lead times and high outbound shipping.

Regional warehouses locate stock closer to demand clusters, dramatically improving delivery responsiveness.

Decentralized regional networks require higher aggregate safety stock across all holding nodes.

Regional setups are advantageous when customer service level expectations demand 24 to 48-hour delivery.

REFERENCE CHAPTER: SUPPLY CHAIN NETWORK STRATEGY - PART 2

Advanced Network Structures: Hybrid Hub-and-Spoke Architectures:

A hybrid network maintains central warehousing for slow-moving, high-value SKUs with high variability. Meanwhile, fast-moving, high-demand items are positioned in regional cross-docks or regional forward nodes. This balances inventory carrying costs through risk pooling while meeting stringent local responsiveness.

Evaluation guideline: In case analysis, both centralized, regional, and hybrid recommendations can be academically valid if backed by sound trade-off analysis of transportation, inventory, and responsiveness.